

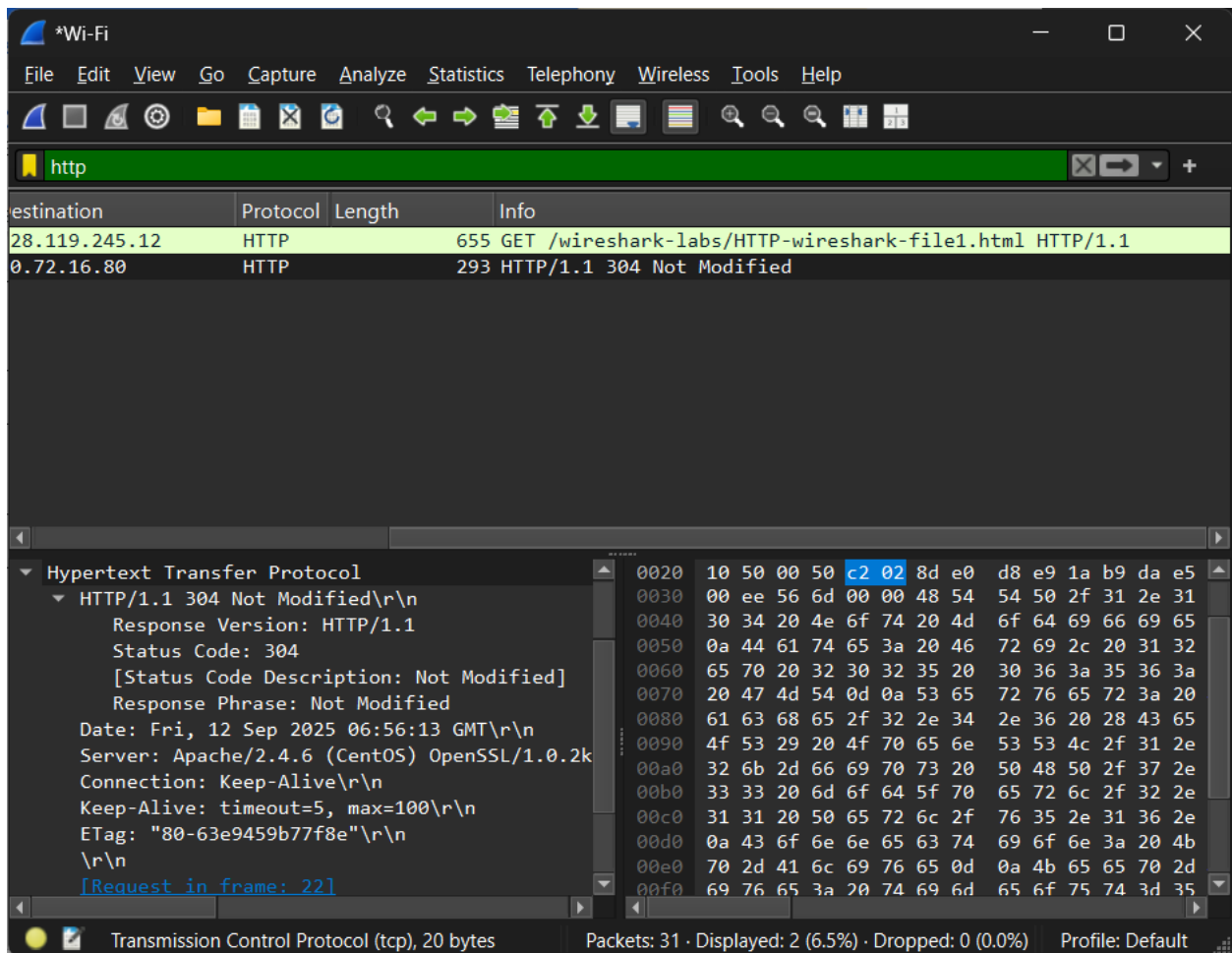
# Worksheet

## Computer Network - Week 2 - HTTP

|              |                       |
|--------------|-----------------------|
| <b>Name</b>  | Farrel Tsaqif Anindyo |
| <b>Class</b> | CSA                   |

## Activity 1 : HTTP GET/Response Interaction

### Screenshots



### Answers

1. Which version of HTTP does your web browser use: 1.0, 1.1, or 2? Which HTTP version is used by the

server?

1.1

2. Which language(s) (if any) can your web browser accept?

En-US (English)

3. What status code does the server return to your web browser?

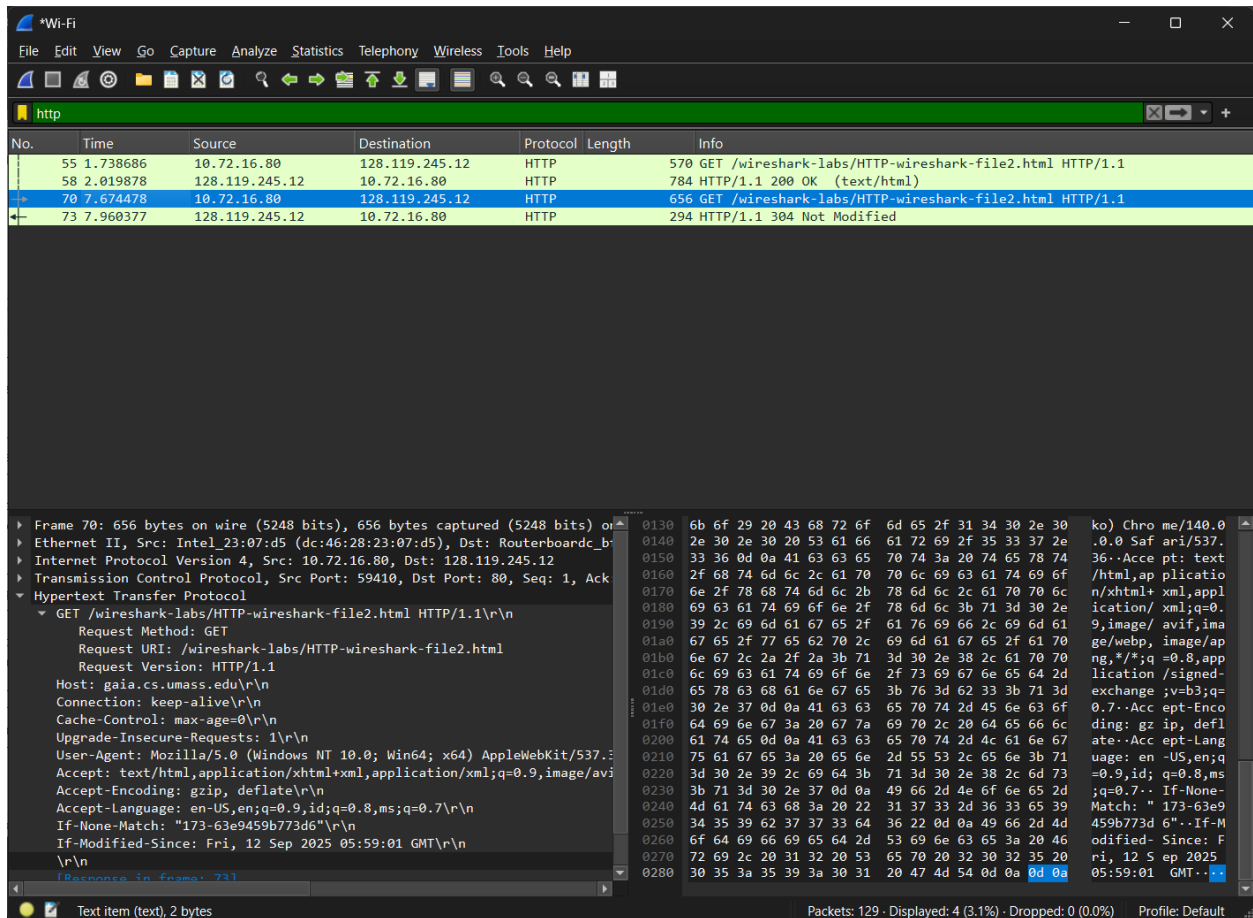
304

4. When was the HTML file you downloaded last modified on the server?

Friday, 12 September 2025

## Activity 2 : Conditional GET/Response Interaction in HTTP

### Screenshots



### Answers

1. Check the first HTTP GET request sent by your browser to the server. Do you see an IF-MODIFIED-SINCE line in this GET request?

No

2. Examine the server's response. Does the server explicitly return the requested HTML file? How can

you tell?

I think it returns the requested HTML file because after GET message there is a name of HTML file

3. Now, check the second HTTP GET request from your browser to the server. Do you see an IF-MODIFIED-SINCE: line in this request? If so, what information follows the IF-MODIFIED-SINCE: header?

Yes, below is the blue text that say, "response in frame: 73" and below it "full request URL: http..."

4. What HTTP status code and phrase does the server return in response to this second HTTP GET? Does the server explicitly return the requested HTML file? Explain.

Status code: 304, I think the server returns the requested HTML file because the HTTP request is sending an appropriate HTTP response back to me

## Activity 3 : Downloading Big Files

### Screenshots

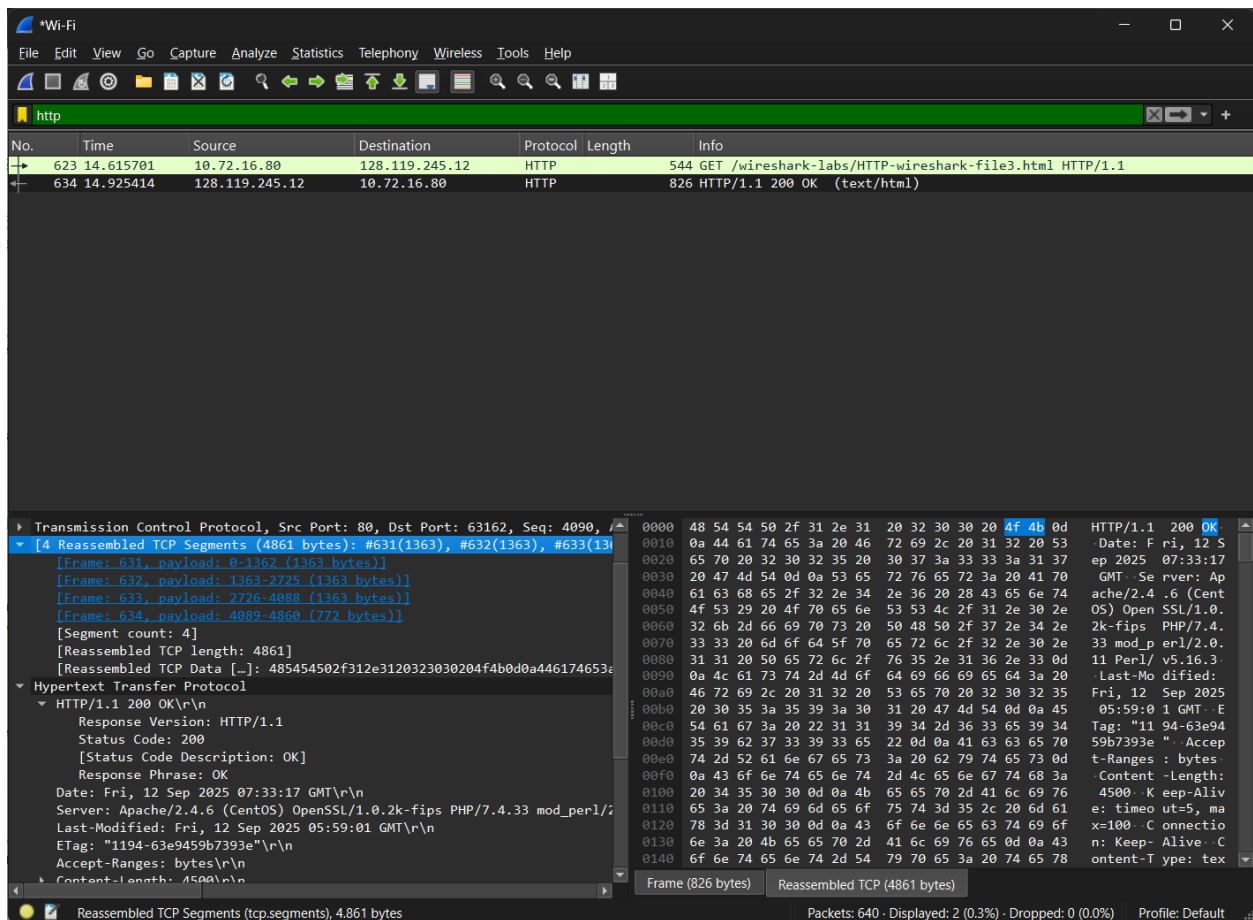
The screenshot displays the Wireshark network traffic capture interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture, and analysis. The packet list pane shows two packets:

| No. | Time     | Source         | Destination    | Protocol | Length | Info                                                   |
|-----|----------|----------------|----------------|----------|--------|--------------------------------------------------------|
| 64  | 1.241806 | 10.72.16.80    | 128.119.245.12 | HTTP     | 544    | GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1 |
| 68  | 1.535478 | 128.119.245.12 | 10.72.16.80    | HTTP     | 826    | HTTP/1.1 200 OK (text/html)                            |

The packet details pane for the selected packet (No. 64) shows the following structure:

- Frame 64: 544 bytes on wire (4352 bits), 544 bytes captured (4352 bits) on : 0000 18 fd 74 bf e4 f3 dc 46 28 23 07 d5 08 00 45 00 ...t...F (#...E
- Ethernet II, Src: Intel\_23:07:d5 (dc:46:28:23:07:d5), Dst: Routerboardc\_bf: 0010 02 12 ca 19 40 00 80 06 00 00 0a 48 10 50 80 77 ...@...H.P.w
- Internet Protocol Version 4, Src: 10.72.16.80, Dst: 128.119.245.12 0020 f5 0c f5 37 00 50 d7 ac fc 64 45 b1 68 39 50 18 ...7 P...dE.h9P
- Transmission Control Protocol, Src Port: 62775, Dst Port: 80, Seq: 1, Ack: 0030 00 ff 92 20 00 00 47 45 54 20 2f 77 69 72 65 73 ...GE T /wines
- Hypertext Transfer Protocol
  - GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1\r\n0040 68 61 72 6b 2d 6c 61 62 73 2f 48 54 54 50 2d 77 hark-lab s/HTTP-w
  - Request Method: GET0050 69 72 65 73 68 61 72 6b 2d 66 69 6c 65 33 2e 68 reshark -file3.h
  - Request URI: /wireshark-labs/HTTP-wireshark-file3.html0060 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f tml HTTP /1.1..Ho
  - Request Version: HTTP/1.10070 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 st: gaia .cs.umass
  - Host: gaia.cs.umass.edu\r\n0080 73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 74 69 6f s.edu -C connectio
  - Connection: keep-alive\r\n0090 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 n: keep- alive: U
  - Upgrade-Insecure-Requests: 1\r\n00a0 70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d pgrade-I nsecure-
  - Requests: 1. Use00b0 52 65 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65 Requests : 1. Use
  - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.3600c0 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 r-Agent: Mozilla
  - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,00d0 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 /5.0 (Wi ndows NT
  - Accept-Encoding: gzip, deflate\r\n00e0 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 10.0; W in64; x6
  - Accept-Language: en-US,en;q=0.9,id;q=0.8,ms;q=0.7\r\n00f0 34 29 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 4) Apple WebKit/5
  - \r\n0100 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 37.36 (K HTML, li
  - [Response in frame: 68]0110 6b 65 20 47 65 63 6b 6f 29 20 43 68 72 6f 6d 65 ke Gecko ) Chrome
  - [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark0120 2f 31 34 30 2e 30 2e 30 2e 30 20 53 61 66 61 72 /140.0.0 .0 Safar
  - 0130 69 2f 35 33 37 2e 33 36 0d 0a 41 63 63 65 70 74 i/537.36 ..Accept
  - 0140 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c : text/h tml,appl
  - 0150 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d ication/ xhtml+xm
  - 0160 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 6d l,applic ation/xm

The status bar at the bottom indicates: Packets: 78 · Displayed: 2 (2.6%) · Dropped: 0 (0.0%) Profile: Default



## Answers

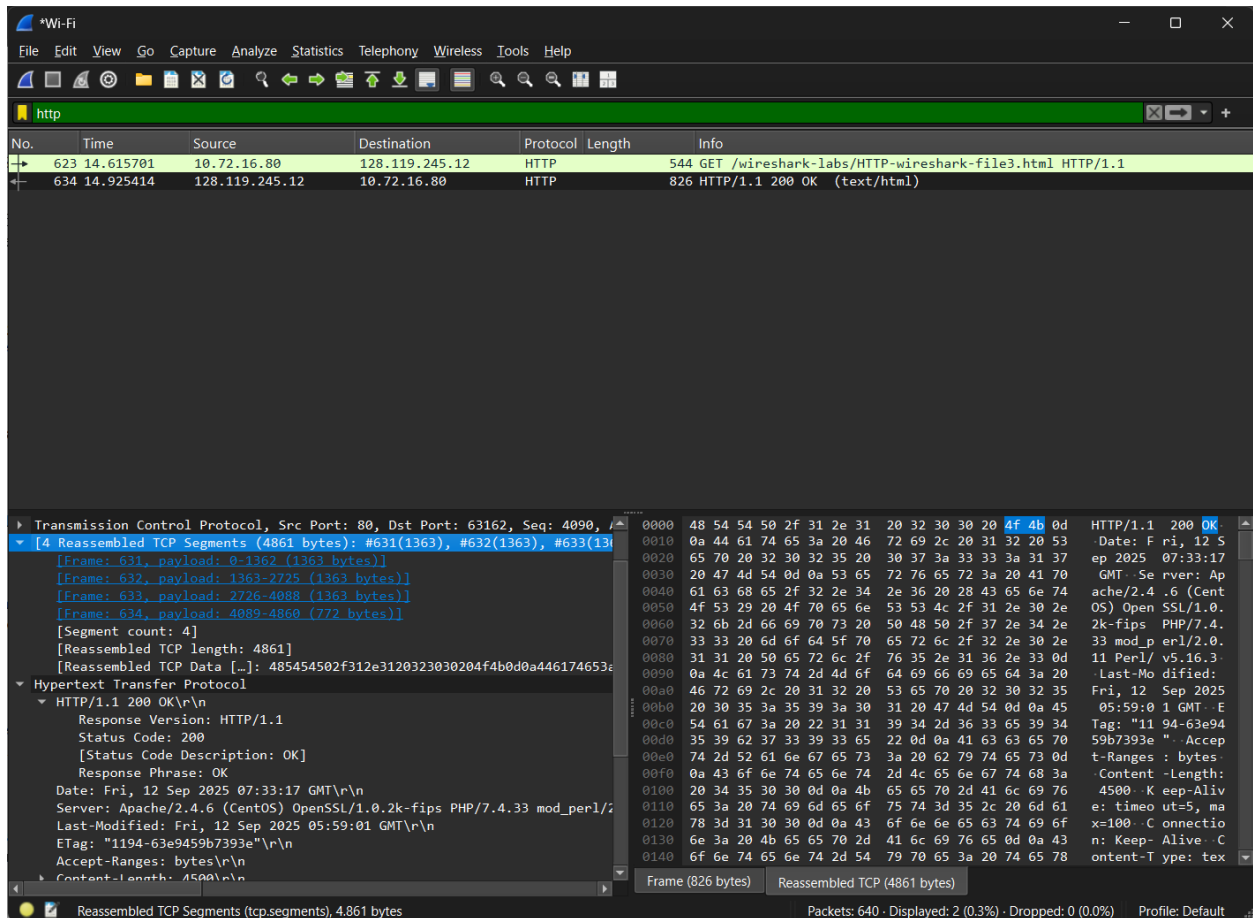
1. How many HTTP GET messages were sent by your browser? Which packet numbers contain these HTTP GET messages?

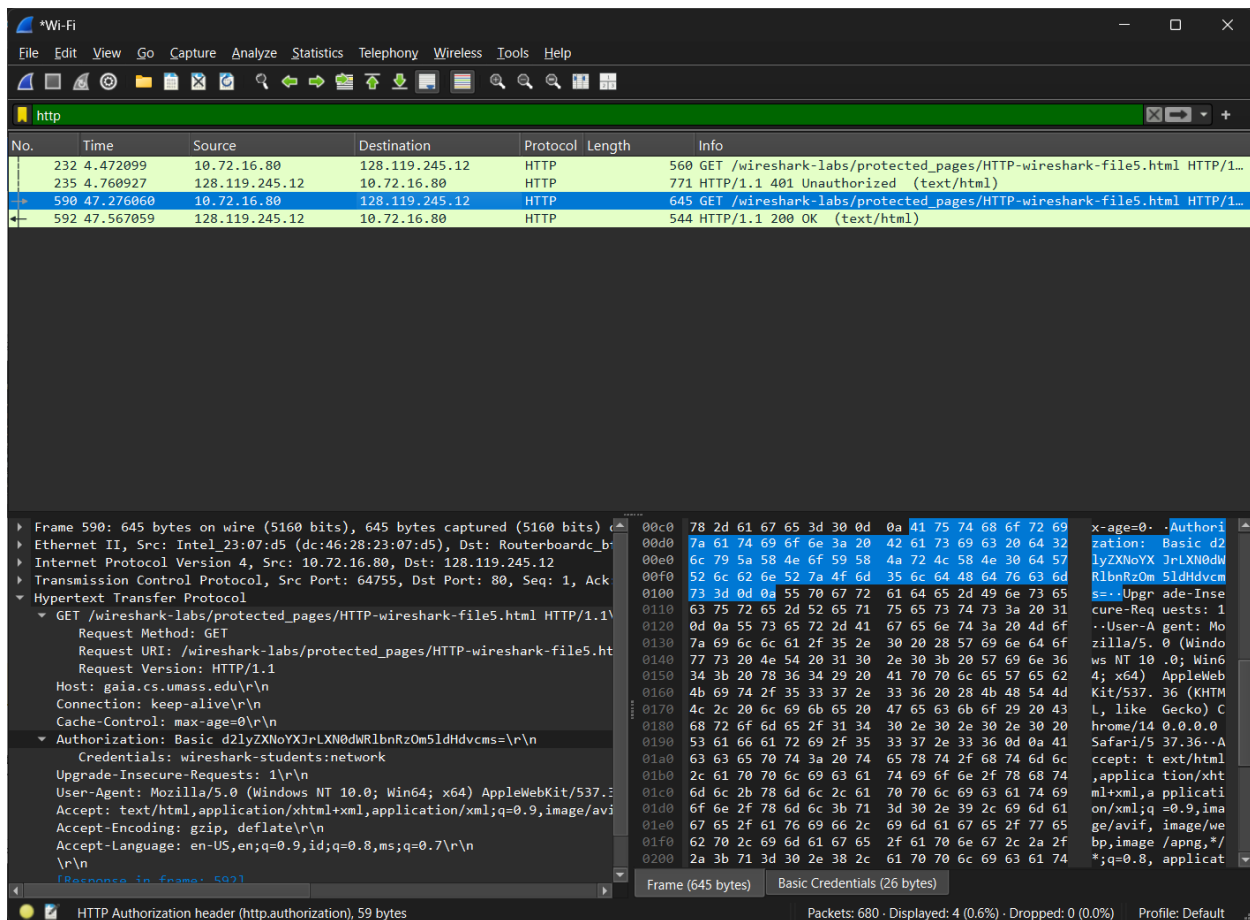
I got one message, packet number: 64

2. How many TCP segments carry the data fragments needed to transmit this large HTML file?

four TCP segments

## Screenshots





## Answers

1. What is the server's response (status code and phrase) to the first HTTP GET message sent by your browser?

Status code: 401, response phrase: unauthorized

2. When your browser sends the HTTP GET message a second time, which new field is included in the GET request?

Authorization line with credential that contain username and password that I input